

Species-Aware Module Review: Bacterial SRP-Dependent Cotranslational Membrane Targeting in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Local bucket reviewed:** KEGG `ppu03060` "Protein export" (19 primary genes) **Module scope:** SRP-dependent cotranslational targeting of nascent membrane proteins — from signal-sequence recognition by the Ffh/SRP-RNA particle, through FtsY capture at the membrane, to the reciprocal GTPase-driven handoff to SecYEG. **Boundary ends at handoff;** SecYEG translocation/insertion, signal-peptide cleavage, and Tat export are downstream/parallel modules.

1. Executive summary

- The SRP-targeting module is **fully satisfiable** in *P. putida* KT2440. All three required components are present in the genome:
- **Ffh** — `ffh` / **PP_1461** (UniProt Q88MV7)
- **SRP RNA** — `ffs` / **PP_mr50** (RefSeq **PP_RS22200**; NCBI GeneID 26969983) — a **non-coding RNA**, confirmed directly in the genome
- **FtsY** — `ftsY` / **PP_5111** (UniProt Q88CR9)
- **Only 2 of the 19** candidate genes in the KEGG `ppu03060` bucket actually belong to this module (`ffh`, `ftsY`). The KEGG "Protein export" bucket is **broader than the module**; the other 17 genes are downstream Sec translocation, YidC insertion, signal peptidases, or the parallel (Sec-independent) Tat pathway.
- The single most important **curation action** is to recognize that a required module component (`ffs` SRP RNA) is a non-coding RNA that is **structurally invisible to a proteome-derived candidate list**. It must be tracked outside UP000000556. It is present (PP_mr50), so the RNA-binding step is **covered**, not a gap.

- **Ffh and FtsY are strictly single-copy** — no paralog ambiguity, no lineage-specific replacement, no over-propagation risk for these two.
- **Evidence caveat:** There is **no direct KT2440 experimental study** of the SRP pathway. All target-organism annotations are computational homology transfer (HAMAP/UniRule, IEA). Transfer from *E. coli*/*B. subtilis* is nonetheless **strong** because SRP/FtsY/SRP-RNA are essential, universally conserved (LUCA-level), and single-copy.

2. Target-organism pathway definition

What is included (module scope): cotranslational recognition of a hydrophobic signal-anchor sequence emerging from the ribosome by bacterial SRP (Ffh + SRP RNA); GTP-dependent capture of the ribosome-nascent-chain (RNC)-SRP complex by the membrane receptor FtsY; reciprocal Ffh↔FtsY GTPase activation; handoff of the RNC to the Sec translocase; and dissociation/recycling of Ffh and FtsY. This corresponds to **GO:0006614 / GO:0006613 "SRP-dependent cotranslational protein targeting to membrane"** (the term already annotated on both Ffh and FtsY in UniProt).

Neighboring processes that must be kept separate (same KEGG bucket, different module):

- **SecYEG protein-conducting channel** (SecY/SecE/SecG) and the **SecA** motor ATPase, **SecB** chaperone, **SecDF-YajC** ancillary complex — this is the *downstream* Sec translocation/insertion machinery. The module boundary explicitly ends at the SRP→SecYEG handoff.
- **YidC** membrane insertase — downstream/parallel insertion (SecYEG-dependent and SecYEG-independent insertion), not part of targeting.
- **LepB** (signal peptidase I) and **LspA** (signal peptidase II) — post-translocation signal-peptide cleavage/maturation.
- **Twin-arginine translocation (Tat)** — a completely separate, *Sec-independent* pathway that exports folded proteins; not SRP-related.

Alternate names / database definitions: KEGG map03060 "Protein export"; the SRP RNA appears as **ffs** (gene), **4.5S RNA** (*E. coli* nomenclature), **small cytoplasmic/SRP RNA**, and **Rfam RF00169** ("Bacteria_small_SRP"). Ffh = "Fifty-four homolog" / SRP54 family; FtsY = SRP receptor (SR / SRα homolog).

3. Expected step model and satisfiability

Module step	Required player	KT2440 locus	Status	Evidence
1a. Nascent signal-sequence binding	Ffh (SRP54 family)	ffh / PP_1461 (Q88MV7)	covered	Homology (HAMAP MF_00306, IEA); single-copy
1b. SRP RNA scaffold binding (7S/4.5S RNA binding)	SRP RNA + Ffh	ffs / PP_mr50 (PP_RS22200) + PP_1461	covered (non-protein component)	Direct genome evidence (NCBI GeneID 26969983); Rfam RF00169
2. Membrane receptor engagement	FtsY (SR family)	ftsY / PP_5111 (Q88CR9)	covered	Homology (HAMAP MF_00920, IEA); single-copy
3. Reciprocal GTPase handoff + recycling	Ffh + FtsY GTPase domains	PP_1461 + PP_5111	covered	SRP54-type GTP-binding domain (PROSITE PS00300) on both

No module step is a gap or "not expected." The module is complete. The only nuance is that step 1b depends on a non-coding RNA that a proteome-only candidate list cannot represent.

4. Candidate genes and evidence (module-relevant genes)

Ffh — **ffh** / PP_1461 / Q88MV7 (458 aa). - Role: binds the hydrophobic signal sequence of the RNC and the SRP RNA; forms SRP; delivers cargo to FtsY; GTPase. - Evidence: ProteinExistence level 3 (inferred from homology); HAMAP-Rule MF_00306; GO GTPase activity, 7S RNA binding, SRP complex, SRP-dependent cotranslational targeting (all IEA). SRP54-type GTP-binding domain (PS00300). - Caveats: EC 3.6.5.4 is the generic SRP-GTPase EC and is fine here. No direct KT2440 data, but transfer is strong (essential, universally conserved, single-copy).

FtsY — **ftsY** / PP_5111 / Q88CR9 (494 aa). - Role: membrane-associated SRP receptor; captures RNC·SRP; reciprocal GTPase; hands RNC to Sec translocase; dissociates. - Evidence: ProteinExistence level 3; HAMAP-Rule MF_00920; GO GTPase activity, signal recognition

particle binding, plasma membrane, SRP-dependent targeting (all IEA). SRP54-type GTP-binding domain (PS00300). - Caveats: Same generic EC 3.6.5.4; appropriate. Note FtsY in *E. coli* lacks the eukaryotic SRα "A-domain" architecture; the KT2440 protein is the standard bacterial one-domain-linker + NG-domain form — no transfer concern.

SRP RNA — `ffs` / PP_mr50 / PP_RS22200 (GeneID 26969983, ~97 nt). - Role: RNA scaffold of SRP; accelerates and catalyzes Ffh–FtsY complex assembly and GTPase activation. - Evidence: **Direct genome annotation** in KT2440 (NCBI Gene); Rfam RF00169. Not in the proteome candidate list because it is non-coding. - Caveat: Must be curated as a non-protein module component.

(The remaining 15 candidate genes — SecY/E/G, SecA/B, SecDF/YajC, YidC, LepB, LspA, and both Tat operons — are annotated correctly but belong to downstream/parallel modules and are out of scope for this module's satisfiability.)

5. Gaps, ambiguities, and likely over-annotations

- **No true gaps** in the SRP-targeting module for KT2440.
 - **Apparent gap that is actually a metadata artifact:** the SRP RNA (`ffs`) is missing from the candidate list only because the list is proteome-derived. Resolved — the locus exists (PP_mr50).
 - **Bucket-level over-inclusion (the main curation issue):** KEGG `ppu03060` conflates ≥ 5 distinct processes (SRP targeting, Sec translocation, YidC insertion, signal-peptide cleavage, Tat export). Scoring all 19 genes as "module coverage" would badly over-count. Only `ffh` + `ftsY` (+ `ffs`) count toward *this* module.
 - **Paralog notes (informational, out of module scope):** KT2440 carries a **duplicated Tat system** (tatABC-I at PP_1039–1041 and tatABC-II at PP_5016–5018) — a genuine *Pseudomonas* feature, but irrelevant to SRP. Ffh and FtsY themselves have **no paralogs**.
 - **No over-propagated SRP annotations detected:** the `ffh` / `ftsY` calls are specific, rule-based, and unambiguous.
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6. Module and GO-curation recommendations

Per-step disposition: - Step 1a (Ffh signal binding): **covered** → PP_1461. - Step 1b (SRP-RNA binding): **covered** → non-protein locus `ffs`/PP_mr50; add an explicit note that this module requires a non-coding-RNA component tracked outside UP000000556. - Step 2 (FtsY engagement): **covered** → PP_5111. - Step 3 (reciprocal GTPase handoff/recycling): **covered** → PP_1461 + PP_5111 (GTPase domains).

Module-level: - Mark the SRP-targeting module **covered/satisfied** for KT2440. - **Do not** treat KEGG `ppu03060` as coextensive with this module. Recommend either (a) splitting the bucket in local metadata into SRP-targeting / Sec-translocation / YidC / signal-peptidase / Tat sub-modules, or (b) tagging the 17 non-SRP genes as "adjacent, downstream/parallel" so they are not scored as SRP-module coverage. - **GO curation:** GO:0006614/GO:0006613 is already correctly applied to Ffh and FtsY. Consider adding an SRP-RNA-linked annotation for `ffs` if the local model supports ncRNA participants. No new GO term request appears necessary. - **New module documents:** not required for SRP itself; a small note documenting the non-protein `ffs` dependency would prevent this false-gap from recurring in other proteome-only reviews.

7. Genes to promote to full `fetch-gene` review

- `ffh` / PP_1461 and `ftsY` / PP_5111 — promote both to full review to formally attach the module-satisfiability call and record that evidence is homology-based (level 3). Low risk, high value for closing the module.
 - `ffs` / PP_mr50 (PP_RS22200) — flag as a **non-protein locus** requiring a curation path that can register a ncRNA participant; this is the one item most likely to be mishandled by protein-centric tooling.
 - The 17 non-SRP genes do **not** need promotion under this module (they belong to their own modules).
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8. Evidence and open questions

Supported by direct target-organism evidence: presence of all three loci (`ffh` PP_1461, `ffs` PP_mr50, `ftsY` PP_5111) in the KT2440 genome/proteome; single-copy status of `ffh` / `ftsY`.

Inferred (homology / rules / neighboring organisms): the biochemical mechanism (signal recognition → FtsY capture → reciprocal GTPase activation → SecYEG handoff → dissociation) is established in *E. coli*, *B. subtilis*, and *T. aquaticus* (PMIDs 15383838, 15680975, 19174514, 30146170, 35065991) and transferred to KT2440. Essentiality is inferred from other bacteria (e.g., SRP disruption impairs growth in *Lactococcus lactis*,

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ffs is essential in *E. coli*,

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Open questions / experiments to resolve gaps: 1. Direct confirmation that ffh, ftsY, and ffs are essential in KT2440 (targeted CRISPRi or a genome-wide essentiality/Tn-seq dataset). No KT2440-specific SRP essentiality study currently exists. 2. Whether *P. putida* SRP substrate range or regulation (e.g., (p)ppGpp inhibition of SRP assembly, shown in *B. subtilis*/*E. coli*,

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behaves as in model organisms — relevant to stress physiology but outside module satisfiability.

Key references

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- Wang et al., SRP-pathway disruption phenotypes in *L. lactis*, 2025
([P 39378800](#)).
- Resources: UniProt Q88MV7 (Ffh), Q88CR9 (FtsY); HAMAP MF_00306, MF_00920; PROSITE PS00300; NCBI Gene 26969983 (`ffs`), PP_mr50/PP_RS22200); Rfam RF00169.